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THE MAGNETIZATION OF THE GREAT WHIN SILL, NORTHERN ENGLAND

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A palaeomagnetic study of samples from 8 localities along the Great Whin Sill (Upper Carboniferous) are presented. Alternating field and thermal demagnetization studies reveal secondary components of both low and high stability. The latter component, which may be as stable as the supposed original remanence, is considered to have been imposed in a normal geomagnetic field, in the opposite polarity to that acting during magma emplacement. The origin of the stable secondary component appears to be reasonably explained by chemical processes as there are evidences of hydrothermal activity in northern England up to at least the Middle Mesozoic. Acceptable palaeomagnetic data have been obtained from 5 sites. Neither the distribution of acceptable palaeomagnetic directions nor the overall mean direction are in agreement with those of a previous study. The estimated pole position in present grids is 44.3°N, 160.8°E.

1. Introduction

In a previous palaeomagnetic survey of the Great Whin Sill (CREER et al., 1959) it was found that the site mean directions of natural remanent magnetization (NRM) had an oval distribution, the variation in the direction of inclination being about twice that in the direction of declination. As the rocks were considered magnetically stable, it was suggested that the distribution of site directions (site means considered as spot readings of the geomagnetic field) represents a secular variation feature. Alternatively, the oval distribution might well, as also discussed by Creer et al., be due to superimposed secondary magnetizations. Unfortunately, no relevant experimental procedure was originally applied in testing the latter hypothesis. Therefore, the object of the present investigation was mainly that of clarifying whether the original distribution was a primary one or if it might simply be due to the disturbing effects from later remagnetization processes.

2. Relevant geological information

The Whin Sill is a collective name of the quartz-dolerite intrusions underlying a large part of northern England. The dolerite sheets are closely related with a large number of WSW and E–W striking dikes (in part considered as feeder dikes), together representing a single and well defined petrographic unit. The average thickness of the intrusion is about 30 m.

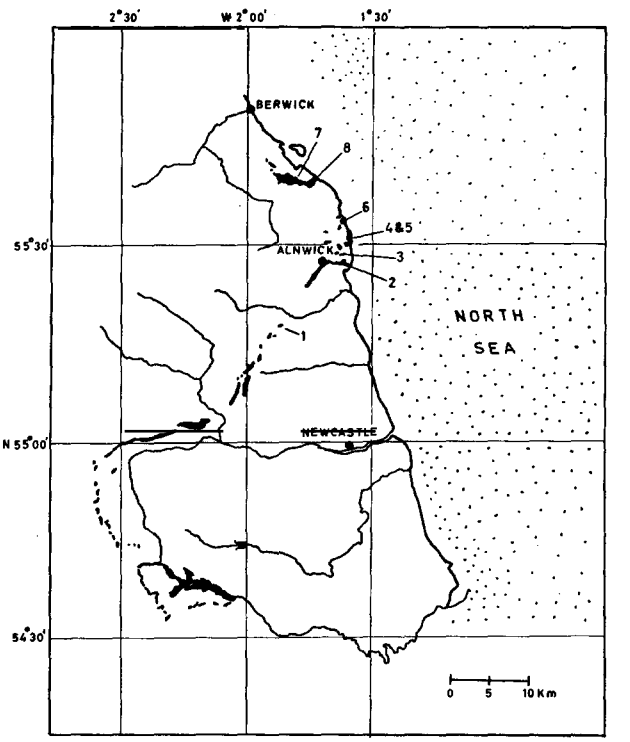


Fig. 1. Map showing the Whin Sill exposures and the distribution of sampling sites.

main occurrence being in the lower part of the succession, the Limestone series, but related dikes cut the Upper Carboniferous Coal Measures (TROTTER and HOLLINGWORTH (1932)). Recent K/Ar age determinations (FITCH and MILLER (1967)) suggest an Upper Carboniferous emplacement 295 ± 6 m.y. After consolidation gas phases produced active hydrothermal solutions which caused deuteric alterations throughout the whole rock formation (WAGER (1929), DUNHAM (1948)), locally converting the original dolerite material into aggregates of carbonates, kaolinite, micaceous clay minerals etc., Following FITCH and MILLER (1967) and DUNHAM et al. (1968) there has been a number of subsequent metasomatic events in northern England, an important phase occurring as late as 170 m.y. ago.

The depth of burial of the Whin Sill has varied considerably but in Northumberland where the present rocks were collected it amounts to 1–1,5 km.

FITCH and MILLER (1967) conclude that the departure of the Whin Sill from a horizontal attitude is due to post-emplacement tilting. In the sites concerned the attitude of the intrusion is roughly horizontal; in only one locality, Wards Hill, the dip was of sufficient magnitude and constancy to give a meaningful measurement.

Using an ordinary geological orientation method altogether 46 oriented hand samples from 8 sites were

3. Laboratory investigations

3.1. THE NATURAL REMANENT MAGNETISM (NRM

On average about 4 cylindrical specimens (h and diameter equal 1.9 cm) were prepared from sample. The intensity of magnetization ranged be 0.35×10^{-3} gauss and 5.1×10^{-3} gauss, the arith mean being 3.04×10^{-3} gauss. Depending o intensity of magnetization short or long period a magnetometers were used.

The specimen directions as well as the site directions have a smeared distribution, both givi overall mean direction of about $D = 186°$, $I =$ in broad agreement with that found by Creer a workers.

The specimen directions are shown in fig. 2. figures stereonets are used throughout. Close open circles represent directions of positive and ne inclination respectively.

3.2. ALTERNATING FIELD ANALYSIS

The a.c. equipment used is similar in construct that applied by CREER (1959). In general, anal two specimens from each sample has been perfc

Different types of directional behaviour stepwise application of a.c. fields of increasing tude are demonstrated in fig. 3. In the majo

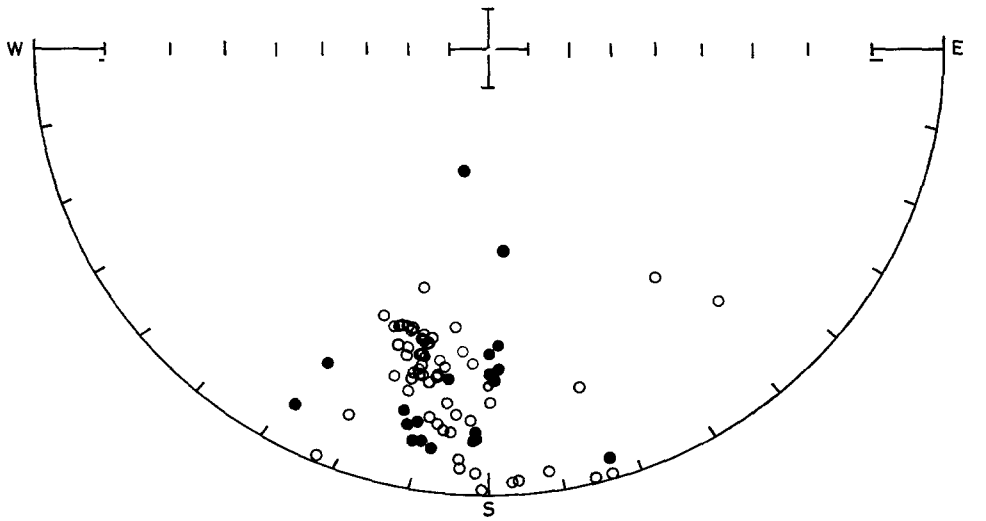


Fig. 2. Directions of natural remanent magnetization of all specimens who have been subjected to stability tes

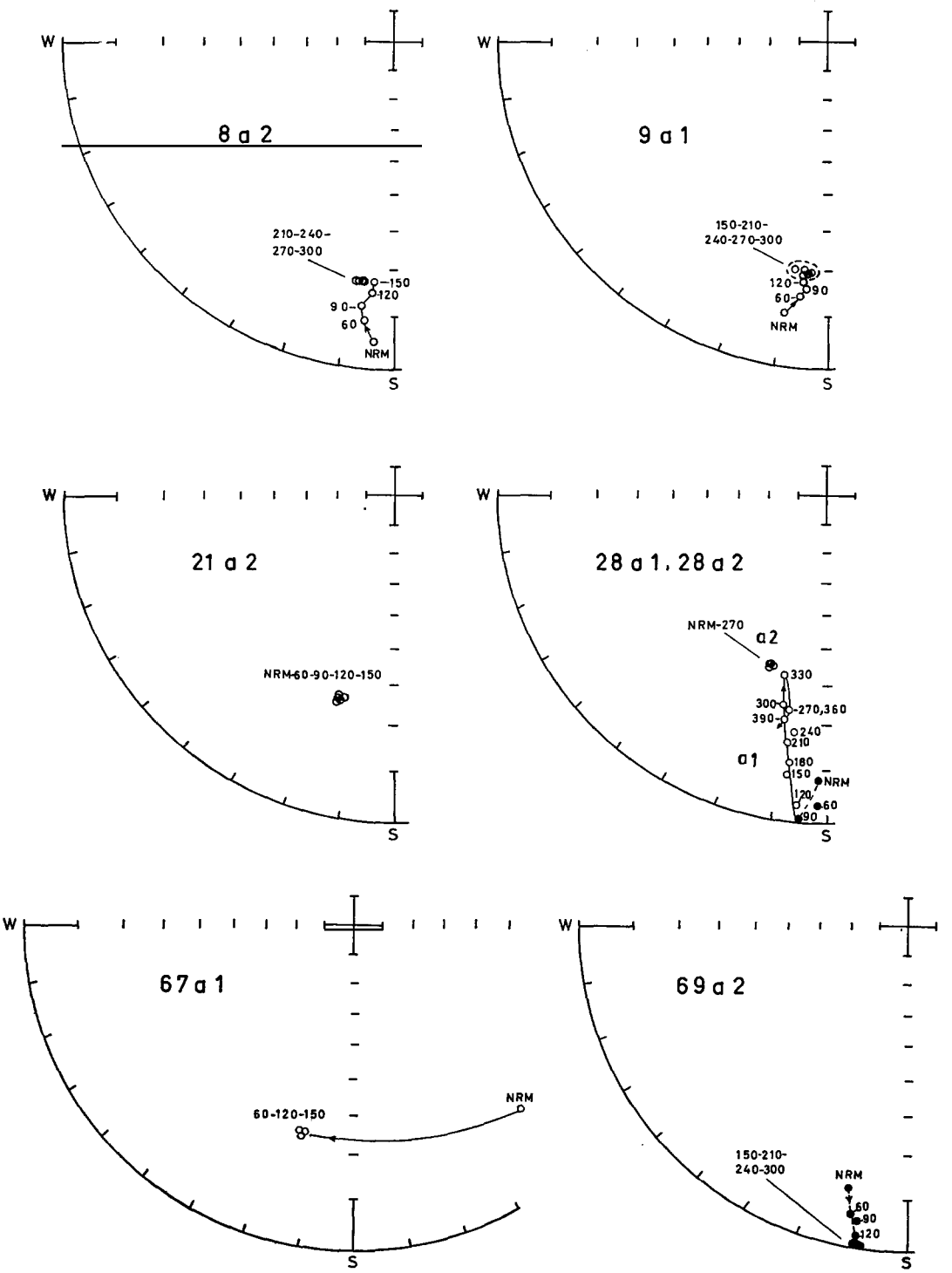


Fig. 3. Specimen directions as a function of increasing alternating field.

specimens investigated there is a well defined directional trend: a movement along a great circle away from the direction of the present local geomagnetic field (cf. 8a2, 9a1, 28a1 and 69a2). Normally the direction changes are less than 15° but sometimes angular changes of about 90° are recorded. In general the low stability component, probably viscous magnetism (VRM), are eliminated after treatment in fields of less than 150 oersteds, but for a number of specimens fields above 300 oersteds are required. However, in the latter case there were a few specimens for which the trend became disrupted before a stable end point had been experimentally verified. This is a well known observation in palaeomagnetic laboratories and instrumental imperfection is likely to be responsible. Specimens showing this behaviour have been excluded from further palaeomagnetic consideration (cf. 28a1).

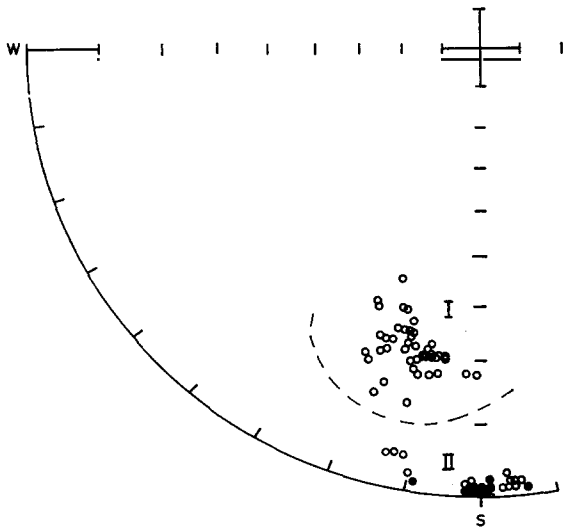


Fig. 4. Magnetization directions after a.c. cleaning of low stability components.

The stable direction of each specimen is represented by a number of closely grouped points of which the mean direction is provisionally adopted as a palaeomagnetic direction. All specimen directions derived in this way are given in fig. 4. As shown, most of these are clustered in two well defined groups, called I and II for reference. Five specimens have an intermediate position between the two groups but they tend to behave as group II when termally demagnetized (see below).

There is some difference in the intensity decay curves of the two groups. Thus, in group I there is, in general, more than 90% of remanence intensity left after treatment in 150 oersteds, while the corresponding intensity of group II specimens varied between 10 and 70% of the original.

3.3. Thermal demagnetization

The palaeomagnetic reliability of the two groups of directions resulting from a.c. treatment were further investigated by thermal demagnetization, the experimental procedure being largely as previously reported (Storetvedt et al. (1968)). However, an application of two Helmholtz coil systems, one of them surrounding the demagnetization equipment and the other serving as a field control arrangement, connected in series, has given a considerable better field cancellation during the thermal process than that previously achieved (Hummervoll (1969)). The "control coil" is equipped with field probes so that field variations caused by changes of coil currents and of geomagnetic field can be adjusted. Although a manual operation has been utilized, the field in the centre of furnace has in general been kept within 5 gammas throughout the heating runs. Unfortunately the new "furnace coil" constructed

Table 1

Account of rock collection after a.c. treatment

Site	Sample nos.	Group I	Group II	Remarks
1 Wards Hill	1– 7		1– 7	
2 Sneableazes	8–14	8–14		
3 Howlet Hill	15–20	15–20		
4 Cullernose Point	21–25	21–25		
5 Whin dike, Cullernose Point	26–30	26–28		nos. 29 and 30, too weakly magnetized for reliable measurement.
6 Snoke Point	31–35			no stable directions obtained non-uniformity of magnetization.
7 Belford quarry	36–40		37–40	no. 36 strongly non-uniformly magnetized.
8 Bamburgh Castle	65–70	65, 67, 70	68, 69	no. 66 broken during cutting.

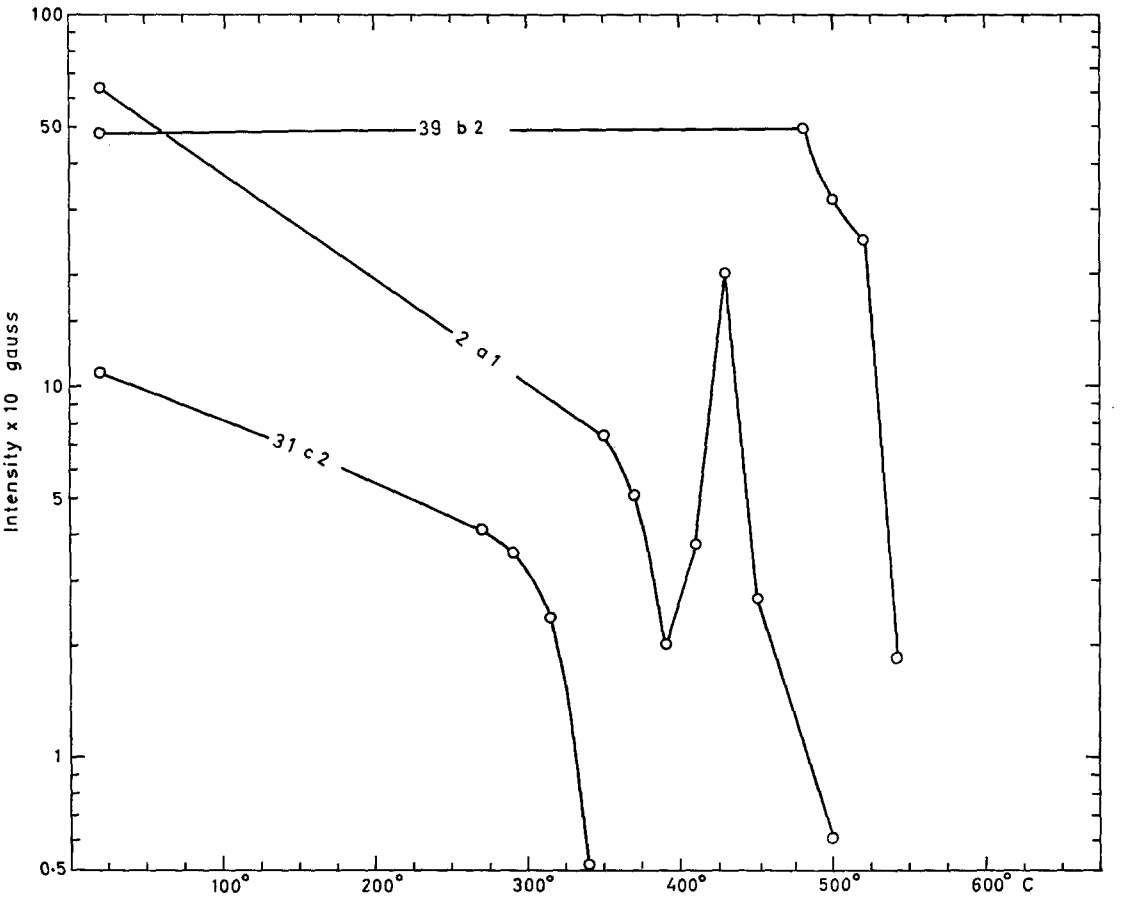


Fig. 5. Representative examples of intensity versus temperature curves.

for this particular device has not proved entirely satisfactory with respect to field gradients. Thus, the specimens in each heating run have been surrounded by stray fields ranging upwards to about 15 gammas.

Different intensity and directional patterns versus temperature are shown in figs. 5 and 6 respectively.

A large number of specimens, in particular those of group I, exhibit a rectangular demagnetization curve, the steep slope occurring in narrow temperature ranges within the interval 500–550 °C. In other cases, i.e. in particular specimens of group II, the blockage temperatures are more distributed and the intensity of magnetization tend to approach zero at lower temperatures (cf. 2a1 and 31c2, fig. 5). On the other hand high field studies reveal indistinguishable thermomagnetic curves for specimens of the two groups, showing Curie temperatures around 550 °C.

In addition to the rapid intensity reduction often encountered, making a rigorous directional analysis very problematic, another general experimental difficulty is the scatter in the results introduced at remanence intensities in the lower range of 10^{-4} gauss. A representative example is specimen 2a1 (cf. figs. 5 and 6) where the direction of magnetization initiates an irregular oscillation above 370 °C. Also intensity fluctuations are generally accompanying the anomalous directional behaviour. A clue to this problem is probably provided by the following observation:

In a heating run to 290 °C an additional field of about 20 gammas was imposed. Thus the total field applied to the specimens during heating and subsequent cooling increased to about 30 gammas. That fields of this magnitude are critical was clearly demonstrated by the results obtained. Thus in 5 of 6 specimens concerned the results are anomalously affected at this temperature, the effect being more marked in the most

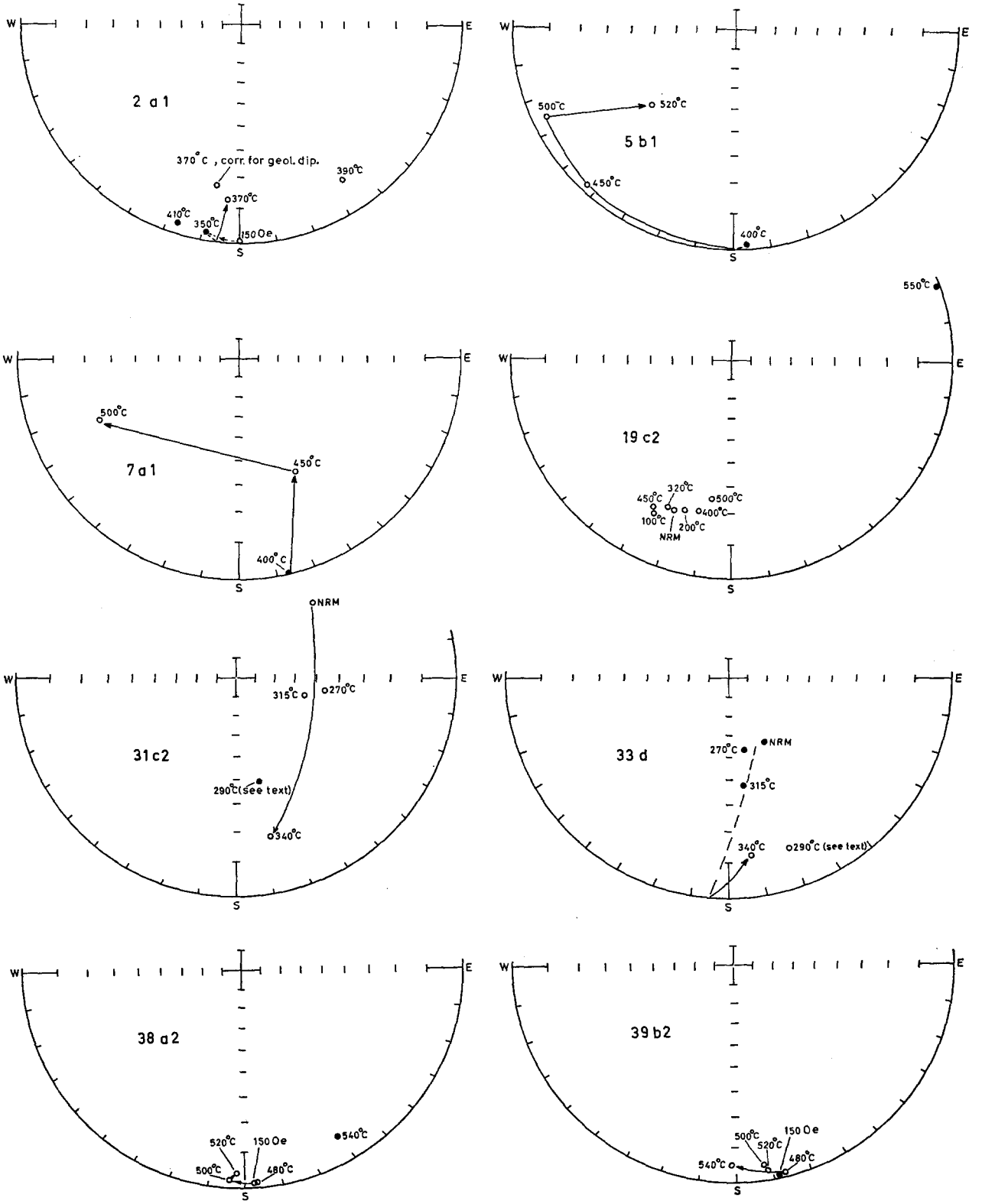


Fig. 6a. Specimen directions as a function of temperature.

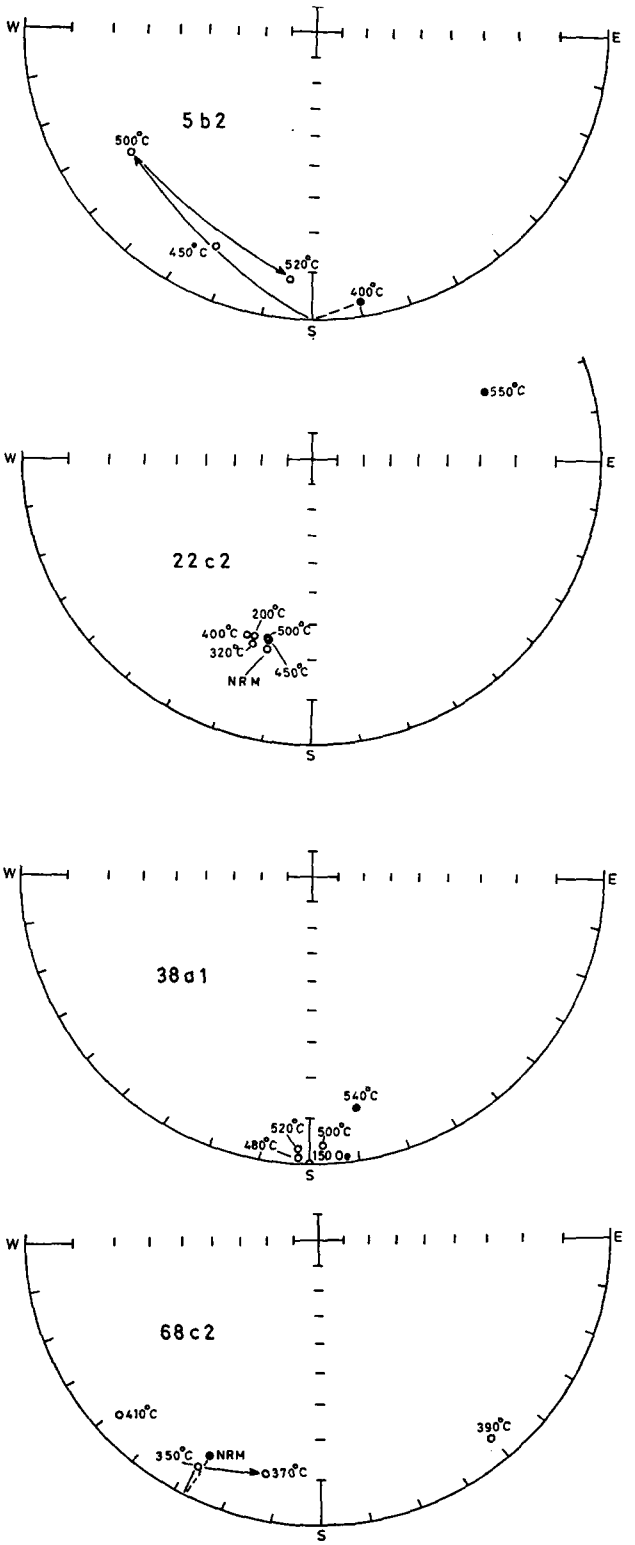


Fig. 6b. Specimen directions as a function of temperature (continued).

weakly magnetized specimens. In particular, specimens of 10^{-4} intensities acquired a strong spurious component of magnetization. On further demagnetization under better field conditions the magnetization directions returned into a well defined pattern (cf. 31c2, 33d, fig. 6).

Although this observation was highly surprising it clearly demonstrated the importance of having the magnetic field ideally cancelled during thermal demagnetization experiments. Nevertheless, the scatter in directions and in intensities appearing at the intensity level of 10^{-4} gauss seems reasonably explained by stray moments acquired during the thermal process. On the other hand it has been possible in some cases to reduce the remanence intensity of stable specimens below the range of 10^{-6} gauss and still obtaining meaningful results. This indicates a great variation in the ability of acquiring TRM in weak fields.

When thermal demagnetization is applied a certain low stability component may exist below 200 °C. The stable direction of samples giving group I on a.c. demagnetization (cf. 19c2 and 22c2, fig. 6) is in very good agreement with those obtained by the a.c. method, the only difference being that the thermal results defining the stable direction are somewhat more scattered than the corresponding a.c. data. Contrary to this, the specimens of group II (cf. 2a1, 5b1, 5b2, 7a1, 38a1, 38a2, 39b2 and 68c2, fig. 6) do not exhibit a stable high temperature direction which is easily accessible. Thus in the majority of these specimens there appears to be direction change towards group I but the directional patterns are always collapsing before a stable end point is obtained. This observation is reasonably understood by the fact that the high temperature component of these specimens appears to be associated with intensities of the order of 10^{-5} gauss, i.e. well below the critical intensity level previously discussed.

4. Evaluation of experimental data

The principal facts extracted from the experiments are:

a. When components of low coercivity (probably of viscous origin) are removed, two groups of remanence directions, called I and II, appear, but

b. subsequent thermal demagnetization studies reveal that the magnetism of the latter group is composite.

However, the most stable component of group II has not been successfully derived, but based on the demagnetization trends there are reasons for suggesting that the high temperature directions coincide with those of group I.

These experimental results probably indicate that the magnetization gained during the original magnetization processes has been successfully deduced in specimens of group I only. The magnetization of group II is considered to be more strongly governed by a stable secondary "overprint". The most likely agencies resulting in secondary components of high thermal stability are thermal and chemical processes. In light of the geological informations available there are reasons for considering the latter one as mainly responsible in the present case. A certain post-Permian CRM-component of normal polarity is likely to deflect the supposed original direction in a manner as recorded (STORETVEDT (1969)). In particular, the remagnetization effect is clearly exhibited by site 8 where the directions (both groups represented) constitute a smeared distribution in the direction of inclination. A CRM-component may be as stable as a thermoremanent magnetization (KOBAYASHI (1959)), whereby the direction of the original resultant may well be the only magnetization direction detected. Problems of this kind seem to be a reasonable explanation of the experimental difficulties in determining the high temperature direction of group II.

As the intrusive formation here concerned has a small vertical extension in addition to having been injected into strata fairly close to the surface, one might suggest that group I carries a TRM rather than a moderate temperature VRM (STORETVEDT (1968)). On the other hand, it is well documented that deuteric alterations were operating after consolidation. These latter processes may well have imposed various CRM-components during times considerably longer than the period of secular variation (these early chemical changes are regarded as having taken place while the geomagnetic field maintained the same polarity as during magma emplacement and must not be confused with the later CRM's mentioned above). On this assumption, the scatter due to secular variation is more likely to be revealed by the dispersion of specimen directions rather than of site mean directions. In fact, it appears rather difficult in general to separate

TABLE 2

Summary of magnetization results from accepted specimens

Site	Specimens	NRM directions	Cleaned directions
2 Sneableazes quarry	8a1	184.5,− 3.7	189.4,−22.3
	a2	184.3,− 4.9	187.5,−17.6
	9a1	188.9,−10.6	186.3,−20.1
	a2	186.9,−12.8	186.6,−20.6
	10b1	188.6,−19.7	190.3,−20.2
	b2	186.8,−25.8	189.9,−20.3
	11b1	183.1,−11.9	180.7,−17.7
	b2	180.5,−14.2	182.6,−18.0
	12a1	182.5,−19.7	187.7,−20.5
	a2	186.5,−21.9	187.8,−20.6
	13a1	193.3,−24.2	193.1,−23.8
	a2	188.7,−17.0	189.0,−17.0
	14b1	195.5,−15.6	196.4,−23.9
	b2	192.6,−14.1	194.0,−23.6
Site mean	N=14	187.3,−15.5	188.6,−20.5
3 Howlet Hill quarry	15a1	189.6,−16.3	191.0,−16.9
	a2	191.7,−16.9	192.7,−19.1
	16a1	192.4,−20.1	189.2,−21.1
	a2	188.4,−18.7	188.1,−20.4
	17a1	193.3,−16.6	192.0,−17.8
	a2	191.0,−17.0	191.6,−19.3
	18a2	187.8,−10.2	190.0,−21.5
	a3	185.4,−11.8	187.7,−20.4
	19a1	185.4,−22.0	188.2,−20.3
	a2	188.8,−17.9	200.7,−18.1
	20a1	200.6,− 7.8	196.1,−14.4
	a2	202.7,− 1.1	197.1,−12.6
Site mean	N=12	191.5,−14.8	192.1,−18.5
4 Cullernose Point	21a1	193.1,−18.4	197.9,−21.5
	a2	196.1,−23.8	194.1,−26.0
	22a1	191.7,−23.5	193.4,−23.2
	a2	190.9,−24.2	193.6,−25.8
	23a1	196.6,−24.7	196.4,−27.9
	a2	195.0,−25.3	195.6,−27.8
	24a1	198.5,−25.3	199.2,−21.9
	a2	197.1,−21.0	197.4,−20.0
	25a1	192.8,−23.2	193.8,−21.5
	a2	195.2,−20.3	194.2,−20.6
Site mean	N=10	194.7,−23.0	195.8,−23.2
5 Whin dike, Cullernose Point	26a1	199.5,−24.1	202.0,−27.5
	a2	201.4,−25.2	201.5,−26.8
	27b2	188.6,−17.9	192.3,−21.4
	28a2	197.9,−33.4	197.9,−33.4
Site mean	N= 4	197.9,−25.1	198.5,−27.2
8 Bamburgh Castle	65b1	164.8,−14.0	198.3,−19.3
	b2	180.5,−15.9	199.7,−17.2
	67a1	138.2,−16.2	195.1,−23.9
	a2	144.2,−25.7	187.7,−20.7
	70c	176.9,+42.0	191.7,−12.2
Site mean	N= 5	160.4,− 7.3	194.5,−18.7

TABLE 3

Mean formation data

	N	Mean direction	R	k	α_{95}	Pole position
Unit weight on specimen	45	192.6, −21.0	44.75	178	1.6	44.3N, 160.8E
Unit weight on sample	24	192.7, −21.1	23.87	177	2.2	44.3N, 160.7E
Unit weight on site	5	193.9, −21.7	4.98	259	4.8	44.4N, 159.1E

the dispersion due to secular variation from dispersion of other sources.

The scatter of group I is of the order of magnitude to be expected from secular variation, but application of the χ^2-test shows that the magnetization directions are not symmetrically distributed around the overall mean direction (in spite of this the statistical parametres given below are estimated by application of FISHER's (1953) statistics). Therefore, if a circular distribution of magnetization directions should be adopted, the observed deviation might indicate an inadequate time representation. Alternatively, certain post-Permian components may be present even in group I deflecting an original circular distribution.

All magnetization directions of group I are listed in table 2. Overall mean directions based on different hierarchal levels together with corresponding statistical parametres and pole positions are given in table 3. Although the specimens concerned are unevenly distributed between sites, the data based on unit weight for each specimen is, according to the above argumentation, considered as the most correct utilization of the observations. However, in the present case the results are practically identical whatever population scheme is followed.

The geological dip was small and uncertain in all sites giving acceptable palaeomagnetic data and therefore no tectonic corrections have been applied. However, in site 1 (group II), where the attitude of the sill is rather well defined, the directions after partial thermal demagnetization are closer to the directions of group I when corrected for geological dip (cf. 2a1, fig. 6).

The estimated palaeomagnetic pole in present coordinates is 44.3°N, 160.8°E, being in close agreement with some other European data based on lavas of similar age (VAN EVERDINGEN (1960), NIJENHUIS (1961), ROCHE et al. (1962), CORNWELL (1967)). However, it may be argued that lavas which have been extruded in low palaeolatitudes (like those considered) may hold complicated remagnetization problems (STORETVEDT (1969)). Although such problems have been largely overlooked in previous investigations, the consistancy of the polar estimates of the above references suggest that these are at least approximately correct. Despite the uncertainty about the amount of secular variation inherent in the Whin Sill data one may suggest, therefore, that the polar estimate as presented represents a reliable Late Palaeozoic European pole position. Nevertheless, it is hoped that confirmations from suitable igneous intrusions (STORETVEDT (1969)) will be forthcoming.

5. Comparison with previous results

The oval distribution of site mean directions obtained by CREER et al. (1959) led them to postulate that the secular variation of the geomagnetic field may be simulated by imagining the geocentric dipole wobbling in a random fashion around a mean position. In this model the distribution of field directions would be latitude dependent, the standard deviations in the direction of declination and of inclination, dD and dI respectively, being expressed as

$$\mathrm{d}D = \Theta_{63}(1+3\sin^2 B)^{-\frac{1}{2}},$$
$$\mathrm{d}I = 2\Theta_{63}(1+3\sin^2 B)^{-1},$$

where B is the co-latitude and Θ_{63} the circular standard deviation of the dipole oscillations. Thus at low latitudes palaeomagnetic observations recorded at a particular point should exhibit, a marked oval distribution; the form and magnitude of the Whin Sill observations (NRM) was in accord with the theory.

In order to test the stability of remanent magnetization Creer and co-workers selected randomly ten pilot specimens and subjected them to a.c. demagnetization. Eight of these specimens were treated in an alternating field of 280 oersteds without applying lower field strengths. The results are assessed as follows: "Specimens showed differences in direction of magnetization before and after cleaning of 9°, 7°, 5°, 9°, 23°, 35°, 6°,

6°, 5° and 17°. These changes in direction are irregular so that the difference between the mean direction of magnetization of these specimens before and after cleaning is only 3° and is not significant". This conclusion is in striking contrast to the experimental results of the present investigation where systematically directed secondary components have been detected. The reason why the general low stability component has gone unnoticed in the first survey is probably that a too high field has been applied in the demagnetization processes adding random components of magnetization comparable in magnitude to the viscous components already broken down.

Although the present study constitutes a much smaller number of sites than encountered in the investigation of Creer and co-workers it is an important fact that the same type of smeared NRM distribution occurs in both cases, no matter which observation level is considered in the present study. However, on proper experimental treatment this distribution is suggested to be a secondary feature, and the overall mean magnetization direction of Creer et al. and that of the cleaned data of the present investigation differ by about 16° (in inclination). Furthermore, the distribution of poles corresponding to the cleaned data of the present investigation is not circular. Consequently, the secular variation model as proposed by Creer and co-workers is not verified by the available Whin Sill data.

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